

QMC Kernel-Bank Approach Report

4 March 2026

Approach

1. QMC-configured kernel pipeline

The QMC setup configures the kernel bank with:

- Sampling method: `qmc` (Sobol low-discrepancy sampling),
- Seed: 42,
- Scrambling: enabled.

Core pipeline settings used in the compared runs:

- Patch setup: patch size 128, inside/outside sampling 40/10 per image.
- Kernel bank: 9 families, 500 kernels/family (4500 total), kernel size 31.
- Selection: MMR, `topM=3250`, selected $K = 90$ kernels.
- Classifier: logistic, epochs = 60, batch size = 64, learning rate = 5×10^{-4} .
- Composite stage: 8 composite kernels built from 1250 selected-source kernels (uniform weights, L1 normalization).

Run Summary

Method comparison is performed between two runs with the same MMR/composite settings:

- **Random sampling baseline.**
- **QMC sampling method.**
- Both runs use: 4500 kernels, MMR selection with $K = 90$, and 8 composite kernels.

Results

2. Test-set method comparison

Evaluation (Test only)	Random sampling	QMC sampling	$\Delta(\text{QMC} - \text{Random})$
Base model AUC	0.919535	0.938699	+0.019164
Base model ACC	0.864571	0.886631	+0.022060
Composite model AUC	0.885451	0.893402	+0.007952
Composite model ACC	0.806286	0.817647	+0.011361

Interpretation:

- On the available test results, QMC improves both AUC and ACC for the base and composite models relative to random sampling.
- The strongest gain appears in the base model AUC (+0.0192 absolute).

Classifier and Kernel Visualizations

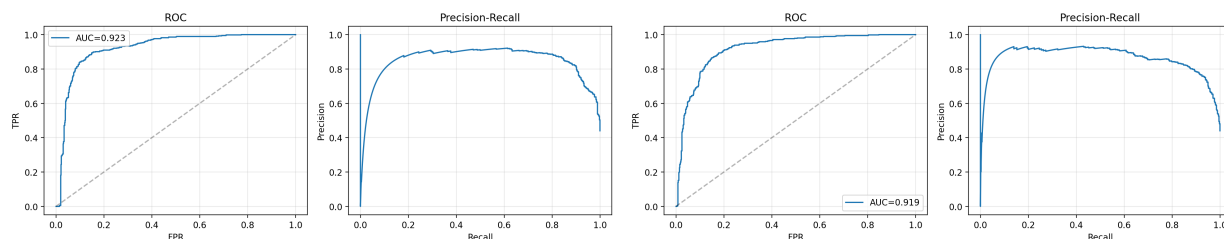


Figure 1: Left: ROC/PR for selected-kernel base classifier. Right: ROC/PR for composite classifier.

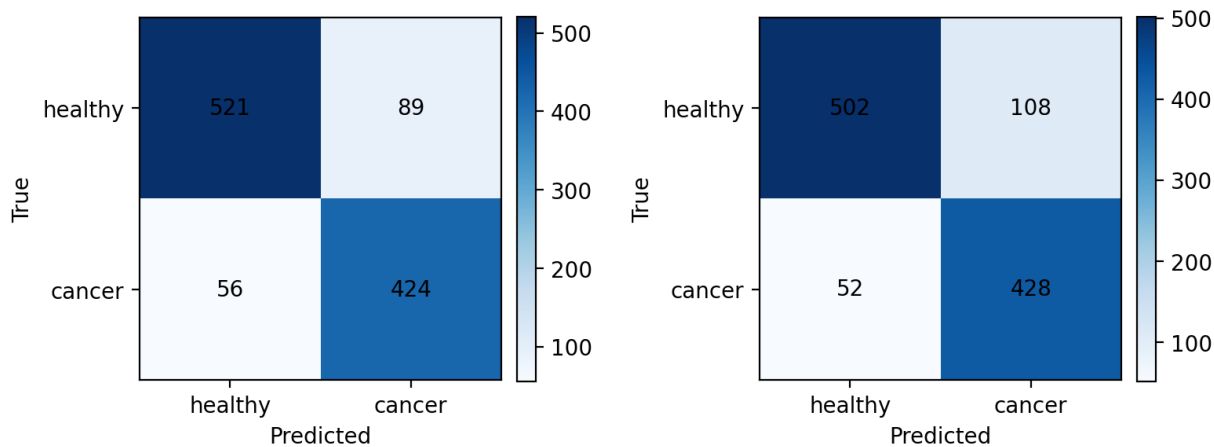


Figure 2: Confusion matrices for base (left) and composite (right) classifiers.

Full-Image Inference + Heatmaps

The notebook generates heatmaps for IMG015, IMG044, IMG059, and IMG330 using the composite model and sliding-window aggregation.

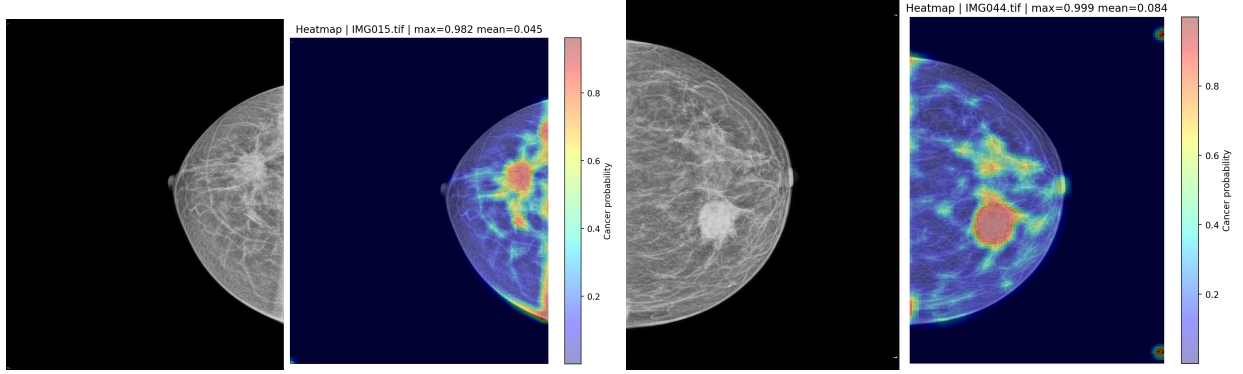


Figure 3: Composite-kernel heatmap pairs (set 1). Left image in each pair is original, right is overlay heatmap.

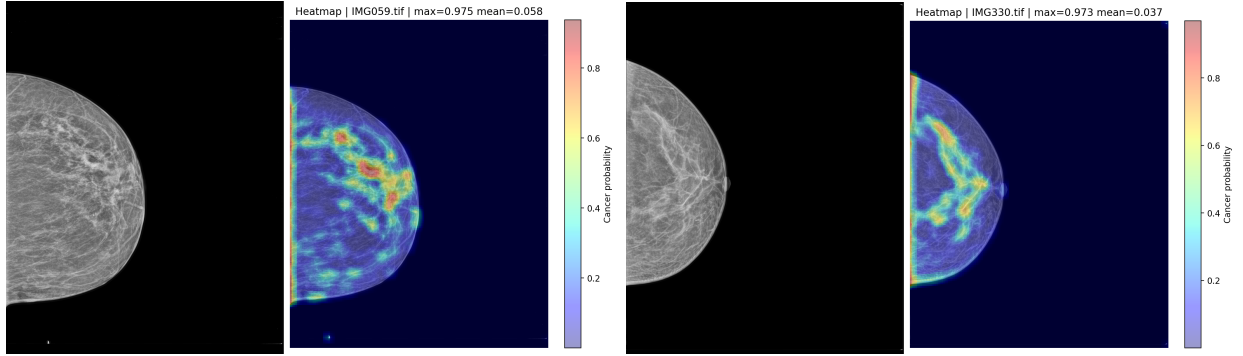


Figure 4: Composite-kernel heatmap pairs (set 2).

Findings

- QMC sampling outperforms random sampling on all reported test metrics in this comparison.
- The gain is largest for the base classifier AUC, and positive but smaller for the composite classifier.

Next Steps

- Combine QMC and composite-kernel training by training composite kernels on data with f_i from 1 to 10.

Conclusion

In this report comparison, QMC is better than random sampling on all reported test metrics (base and composite, AUC and ACC). The evidence supports QMC as the preferred sampling method for this kernel-bank pipeline.